C12.1 About DVM transactions

DVM transactions support the maintenance of a virtual memory system and are used to pass operations that cannot be conveyed using the normal coherency transactions. Support for Distributed Virtual Memory transactions is a design-time option for a component. Components must either fully participate in the distributed virtual memory scheme or they must never participate. Components that are participating must be capable of receiving any distributed virtual memory transaction and responding appropriately.

A DVM scheme has the following transaction types:

DVM Operation

These transactions convey particular operations, such as a *Translation Lookaside Buffer* (TLB) invalidation request. A component can issue concurrent DVM Operations. See *DVM Operations* on page C12-288 for more information.

DVM Sync This is a synchronization transaction that a component issues to check that all previous DVM Operations that it has issued have completed.

DVM Complete

This transaction is issued in response to a DVM Sync transaction. It is issued by a component that has received a number of DVM Operations followed by a DVM Sync. The DVM Complete indicates that all the required operations and any associated transactions have completed.

DVM Operations can require multiple transactions to convey the required information. In this case, the first transaction provides sufficient information to determine whether another transaction is required.

Note

DVM transactions only operate on read-only structures, such as Instruction cache, Branch Predictor, and TLB, and therefore only invalidation operations are required. The concept of cleaning does not apply to a read-only structure. This means that, it is functionally correct to invalidate more entries than the DVM Operation requires, although this can affect performance.

Virtual memory systems typically use a TLB that retains a copy of recent virtual-physical address translations. When an invalidation of a TLB entry is requested, the DVM Operation must be signaled as complete only when any previous transactions that have been using the invalidated translation have completed.

A component must respond in a protocol compliant manner to all DVM messages, even those that it does not support. This permits components with differing subsets of supported messages to interoperate.

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